SG No. 140 D_{4h}^{18} $I4/mcm$ [tetragonal]

* plus set: $+[0,\ 0,\ 0]$, $+\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$

Table 1: Wyckoff site: `4a`, site symmetry: `422`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,2,3,4,5,6,7,8]`
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[9,10,11,12,13,14,15,16]`

Table 2: Wyckoff site: `4b`, site symmetry: `-42m`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,2,5,6,11,12,15,16]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[3,4,7,8,9,10,13,14]`

Table 3: Wyckoff site: `4c`, site symmetry: `4/m..`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,3,4,9,10,11,12]`
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[5,6,7,8,13,14,15,16]`

Table 4: Wyckoff site: `4d`, site symmetry: `m.mm`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,2,7,8,9,10,15,16]`
2	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[3,4,5,6,11,12,13,14]`

Table 5: Wyckoff site: `8e`, site symmetry: `..2/m`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,7,9,15]`
2	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[2,8,10,16]`
3	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[3,5,11,13]`
4	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[4,6,12,14]`

Table 6: Wyckoff site: `8f`, site symmetry: `4..`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3,4]`
2	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[5,6,7,8]`
3	$[0,\ 0,\ -z]$	`[9,10,11,12]`
4	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[13,14,15,16]`

Table 7: Wyckoff site: `8g`, site symmetry: `2.mm`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,2,15,16]`
2	$\left[\frac{1}{2},\ 0,\ z\right]$	`[3,4,13,14]`
3	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[5,6,11,12]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	`[7,8,9,10]`

Table 8: Wyckoff site: `8h`, site symmetry: `m.2m`

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ 0\right]$	`[1,7,10,16]`
2	$\left[-x,\ \frac{1}{2}-x,\ 0\right]$	`[2,8,9,15]`
3	$\left[\frac{1}{2}-x,\ x,\ 0\right]$	`[3,5,12,14]`
4	$\left[x+\frac{1}{2},\ -x,\ 0\right]$	`[4,6,11,13]`

Table 9: Wyckoff site: `16i`, site symmetry: `..2`

No.	position	mapping
1	$\left[x,\ x,\ \frac{1}{4}\right]$	`[1,7]`
2	$\left[-x,\ -x,\ \frac{1}{4}\right]$	`[2,8]`
3	$\left[-x,\ x,\ \frac{1}{4}\right]$	`[3,5]`
4	$\left[x,\ -x,\ \frac{1}{4}\right]$	`[4,6]`
5	$\left[-x,\ -x,\ \frac{3}{4}\right]$	`[9,15]`
6	$\left[x,\ x,\ \frac{3}{4}\right]$	`[10,16]`
7	$\left[x,\ -x,\ \frac{3}{4}\right]$	`[11,13]`
8	$\left[-x,\ x,\ \frac{3}{4}\right]$	`[12,14]`

Table 10: Wyckoff site: `16j`, site symmetry: `.2.`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[1,6]`
2	$\left[-x,\ 0,\ \frac{1}{4}\right]$	`[2,5]`
3	$\left[0,\ x,\ \frac{1}{4}\right]$	`[3,7]`
4	$\left[0,\ -x,\ \frac{1}{4}\right]$	`[4,8]`
5	$\left[-x,\ 0,\ \frac{3}{4}\right]$	`[9,14]`
6	$\left[x,\ 0,\ \frac{3}{4}\right]$	`[10,13]`
7	$\left[0,\ -x,\ \frac{3}{4}\right]$	`[11,15]`
8	$\left[0,\ x,\ \frac{3}{4}\right]$	`[12,16]`

Table 11: Wyckoff site: `16k`, site symmetry: `m..`

No.	position	mapping
1	$[x,\ y,\ 0]$	`[1,10]`
2	$[-x,\ -y,\ 0]$	`[2,9]`

continued ...

Table 11

No.	position	mapping
3	$[-y,\ x,\ 0]$	`[3,12]`
4	$[y,\ -x,\ 0]$	`[4,11]`
5	$\left[-x,\ y,\ \frac{1}{2}\right]$	`[5,14]`
6	$\left[x,\ -y,\ \frac{1}{2}\right]$	`[6,13]`
7	$\left[y,\ x,\ \frac{1}{2}\right]$	`[7,16]`
8	$\left[-y,\ -x,\ \frac{1}{2}\right]$	`[8,15]`

Table 12: Wyckoff site: `16l`, site symmetry: `..m`

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ z\right]$	`[1,16]`
2	$\left[-x,\ \frac{1}{2}-x,\ z\right]$	`[2,15]`
3	$\left[\frac{1}{2}-x,\ x,\ z\right]$	`[3,14]`
4	$\left[x+\frac{1}{2},\ -x,\ z\right]$	`[4,13]`
5	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[5,12]`
6	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	`[6,11]`
7	$\left[x+\frac{1}{2},\ x,\ \frac{1}{2}-z\right]$	`[7,10]`
8	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{2}-z\right]$	`[8,9]`

Table 13: Wyckoff site: `32m`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[-x,\ -y,\ z]$	`[2]`
3	$[-y,\ x,\ z]$	`[3]`
4	$[y,\ -x,\ z]$	`[4]`
5	$\left[-x,\ y,\ \frac{1}{2}-z\right]$	`[5]`
6	$\left[x,\ -y,\ \frac{1}{2}-z\right]$	`[6]`
7	$\left[y,\ x,\ \frac{1}{2}-z\right]$	`[7]`
8	$\left[-y,\ -x,\ \frac{1}{2}-z\right]$	`[8]`
9	$[-x,\ -y,\ -z]$	`[9]`
10	$[x,\ y,\ -z]$	`[10]`
11	$[y,\ -x,\ -z]$	`[11]`
12	$[-y,\ x,\ -z]$	`[12]`
13	$\left[x,\ -y,\ z+\frac{1}{2}\right]$	`[13]`
14	$\left[-x,\ y,\ z+\frac{1}{2}\right]$	`[14]`
15	$\left[-y,\ -x,\ z+\frac{1}{2}\right]$	`[15]`
16	$\left[y,\ x,\ z+\frac{1}{2}\right]$	`[16]`